Nuclear processes in stars

in thermal equilibrium, the amount of energy produced through nuclear processes in the core of a star is exactly balanced by the energy released from the surface through the luminosity of the star.

* energy transport is required though to go from core to surface.

→ nuclear reaction rate sets how fast a star burns through its fuel.

Nuclear processes also convert H, He into heavier elements ⇒ nucleosynthesis.

* H, He were formed in the early universe during recombination when universe cooled enough that radiation didn't knock electrons out of orbits around protons

Properties of nuclear reactions:

$X + a \longrightarrow Y + b$ sometimes written as $X(a,b)Y$

X, Y are nuclei

a, b are particles: either another nucleus, or a γ-photon. a ~ nucleus, b ~ both

It is possible for a reaction to produce more than one particle, especially in weak interactions where bound states are not produced.

→ we will mostly work with strong interactions

Each nucleus is described by

Z_i : charge given by # of protons in the nucleus

A_i : baryon number given by # of protons & neutrons in the nucleus.

→ both Z & A are conserved during a nuclear reaction. Then:

$$Z_X + Z_a = Z_Y + Z_b$$

$$A_X + A_a = A_Y + A_b$$

if a or b are not nuclei, $A_i = 0$

<u>nuclear energy</u>:

nuclei are bound together by the strong nuclear force; this adds energy such that the sum of all the masses of the nucleons is not the mass of the nucleus. This binding energy is defined as

$$E_{B,i} = \left[(A_i - Z_i)m_n + Z_i m_p - m_i\right]c^2$$

m_p: proton mass $\quad$ m_i : mass of nucleus i

m_n : neutron mass

note: $\sum_i A_i = 0$, but $\sum_i M_i \neq 0$!

$\rightarrow$ the net Δm is converted into energy w/ $E = \Delta m c^2$, here

$$\Delta m = m_X + m_a - m_Y - m_b$$, then

the energy released in the nuclear reaction $X(a,b)Y$ is

$$Q = (m_X + m_a - m_Y - m_b)c^2$$

$Q > 0$: exothermic ; $Q < 0$: endothermic

Because $\sum_i Z_i = 0$, the # of electrons is conserved in a nuclear reaction. This means that we can write Δm in terms of the atomic masses rather than the nuclear masses. Then we can write:

$$\Delta M_i = (m_i - A_i m_u)c^2 \equiv \text{mass defect}$$

the nucleon number is also conserved so we can write

$$Q = \Delta M_X + \Delta M_a - \Delta M_Y - \Delta M_b$$

Because each nucleon can be in a different isotopic state, it is more useful to write the binding energy as the binding energy per nucleon: E_B/A

$$\left(\frac{E_B}{A}\right)_{{}^3_2He} : \quad A = 3 \quad Z = 2 \qquad m_{{}^3_2He} = 3.016029 \; m_u$$

$$\frac{E_B}{A} = \frac{\left[(3-2)m_n + 2m_p - m_{{}^3_2He}\right]c^2}{3}$$

$m_p = 1.00727647 \; m_u$

$m_n = 1.008665 \; m_u$

$$m_n + 2m_p - m_{{}^3_2He} = 0.00857747 \; m_u$$

$* \; m_u c^2 = 931.494 \text{ MeV}$

$$\longrightarrow \left(\frac{E_B}{A}\right)_{{}^3_2He} = (0.00286)(931.494 \text{ MeV})$$

$$= 2.663 \text{ MeV}$$

Fast saturation in A up to $E_B/A \sim 8$ MeV

$\longrightarrow$ max is for ${}^{56}Fe$ w/ $\frac{E_B}{A} = 8.79$ MeV show Fig 6.1

Structure in the curve comes from shell structure.

Decrease beyond ^{56}Fe because Z starts to increase & the Coulomb force does not saturate with A.

This peak tells us that energy can be gained from the fusion of lighter elements into heavier elements, because $\frac{E_B}{A}$ increases with each nucleon up to ^{56}Fe.

This is why iron is the final endpoint of stellar evolution.

The vast majority of the energy is released in first step of $H \longrightarrow Fe$ transition with $H \rightarrow He$ releasing 7 MeV!

Thermonuclear reaction rates:

consider: $X(a,b)Y$ ← x is bombarded w/ a that has velocity v

The rate of reactions then depends on the cross section

$$\sigma = \frac{\text{\# of reactions per second}}{\text{flux of a particles}} = \frac{\#/\text{time}}{\#/\text{time}/\text{area}} = \text{area}$$

The number densities are then

$n_x = n_i$; $n_a = n_j$ & the flux is

$$n_j v = \text{incident flux} = [m^{-3}\ \tfrac{m}{s}] = [\tfrac{1}{m^2 s}]$$

$\Rightarrow$ define the rate as

$$r_{ij} = n_i n_j v \sigma \quad \text{if } i \neq j$$

If $i = j$, the possible number of reacting pairs is $\frac{1}{2} n_i (n_i - 1) \simeq \frac{1}{2} n_i^2$; if $n_i \gg 1$.

$$\Rightarrow \quad r_{ij} = \frac{1}{1+\delta_{ij}} n_i n_j v \sigma$$

σ depends on the relative velocity between the particles.

In an ideal gas in local thermodynamic equilibrium, the velocity is distributed by the Maxwell-Boltzmann distribution:

$$\phi(v) = 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT}$$

here: $m = \dfrac{m_i m_j}{m_i + m_j}$

The rate of interactions is then

$$r_i r_j = \frac{1}{1+\delta_{ij}} n_i n_j \int_0^\infty dv\, \phi(v)\, \sigma(v)\, v$$

$$= \frac{n_i n_j \langle \sigma v \rangle}{1 + \delta_{ij}}$$

We can rewrite this in terms of kinetic energy using $E = \frac{1}{2} m v$; $\phi(v)\, dv = \phi(E)\, dE$

Then,

$$\langle \sigma v \rangle = \left(\frac{8}{\pi M}\right)^{1/2} (kT)^{-3/2} \int_0^\infty dE\, \sigma(E)\, E\, e^{-E/kT}$$

⟶ this turns the reaction rate into a temperature dependence only.

... as long as we understand $\sigma(E)$!

<u>Nuclear cross sections</u>:

σ : probability of nuclear reaction.

⟶ difficult to obtain!

classically, the cross section is just

$\sigma = \pi (R_i + R_j)^2$ where R_i, R_j are the radii

⟶ this works for big stuff!

for a nucleus:

$$R_i \simeq R_0 A_i^{1/3} \quad \text{w/ } R_0 = 1.44e{-13}\,\text{cm}$$

$$\Rightarrow \sigma \sim 10^{-25} - 10^{-24}\ \text{cm}^2$$

but! $\lambda = \frac{\hbar}{p} = \frac{\hbar}{\sqrt{2mE}}$; m = reduced mass

$\longrightarrow$ this is the radius at which particles know each other

$$\Rightarrow \sigma = \pi \lambda^2 \text{ where } \lambda > R_i + R_j$$

but it's not quite that easy.

(1) The Coulomb force acts at long range

(2) The type of reaction defines the type of force that acts. If it is only nucleons involved (strong force), our λ-dep. σ kind of works. The EM force which is in play for γ-photons, σ is reduced. If an $e^- \bar{\nu}$ or $e^+ \nu$ is emitted, the weak force is in play and σ is reduced further.

(3) resonances $\longrightarrow$ see section on this

Coulomb barrier:

coulomb potential: $V(r) = \frac{Z_i Z_j e^2}{r} = 1.44\, Z_i Z_j \left[\frac{\text{fm}}{r}\right]$ MeV

for the strong nuclear force to take over, the particles need to be within $r \sim R_0 A^{1/3}$, so they must overcome the coulomb barrier $E = V(r) \simeq Z_i Z_j$ MeV

In stellar interiors

$$\langle E \rangle = \frac{3}{2} kT \approx 1.3 \text{ keV} \; @ \; T \sim 10^7 \text{ K}.$$

$$\langle E \rangle < E_{coulomb} * 1000 \; !$$

→ need quantum mechanical tunneling w/ probability

$$P \sim e^{-\int_{r_n}^{r_c} dr \sqrt{2m(V(r)-E)}/\hbar} \quad ; \quad r_c = \frac{Z_i Z_j e^2}{E}$$

$$P = P_0 \, e^{-bE^{-1/2}} \quad \text{where } b = \frac{2\pi Z_i Z_j e^2}{\hbar} \sqrt{\frac{m}{2}}$$

$$\propto Z_i Z_j \left(\frac{A_i A_j}{A_i + A_j} \right)$$

the heavier the nucleus, the higher the energy required, and thus the hotter the temperature needed.

* Show Fig 6.2

Astrophysical cross section:

$$\sigma(E) = S(E)\, P(E) / E$$

↳ lab experiments give us $S(E)$, but they usually occur at higher energies & so we have to extrapolate.

Temperature dependence of reaction rates:

$$\langle \sigma v \rangle = \left(\frac{8}{\pi M}\right)^{1/2} (kT)^{-3/2} \int_0^\infty dE \; S(E) \, e^{-[E/kT + b/E^{1/2}]}$$

* assume $S(E)$ varies slowly w/ E
* this considers the combined effects of the Maxwell-Boltzmann distribution which decreases rapidly with E and the QM tunneling which increases rapidly with E.

⟶ the peak of the integrand is called the Gamow Peak.

We can find the energy of the Gamow Peak by taking $df/dE = 0$ where $f(E) = e^{-[E/kT + b/E^{1/2}]}$

$$\frac{dF}{dE} = -\left[\frac{E}{kT} + \frac{b}{E^{1/2}}\right]\underbrace{\left[\frac{1}{kT} - \frac{1}{2}\frac{b}{E^{3/2}}\right]}_{=0} e^{-[\;\;]} = 0$$

$$E_0 = \left(\frac{1}{2} b kT\right)^{2/3} = 5.665 \left(Z_i^2 Z_j^2 A\right)^{1/3} \left(\frac{T}{10^7 K}\right)^{2/3} \text{keV}$$

$\langle \sigma v \rangle$: increases strongly w/ T

: decreases strongly w/ Coulomb barrier.

*show Fig 6.4

See notes for analytic description & e^- screening.

Energy generation rates & composition changes:
we now have $\langle \sigma v \rangle$, so we can write down the reaction rate and thus the energy generation rate.

Each reaction releases Q_{ij} & thus $Q_{ij} r_{ij}$ is the energy per unit volume per second. We can write this as the energy per volume per time per mass as

$$\epsilon_{ij} = \frac{Q_{ij} r_{ij}}{\rho}$$

If we plug in r_{ij} we get

$$\epsilon_{ij} = \frac{Q_{ij} n_i n_j \langle \sigma v \rangle}{(1+\delta_{ij})\rho} \quad \text{where} \quad n_i = \frac{X_i \rho}{A_i m_u}, \text{ then}$$

$$\epsilon_{ij} = \frac{Q_{ij}}{(1+\delta_{ij}) A_i A_j m_u^2} \rho X_i X_j \langle \sigma v \rangle$$

using the reduced mass: $A = \frac{A_i A_j}{A_i + A_j}$ and

$$q_{ij} = \frac{Q_{ij}}{m_i + m_j} \cong \frac{Q_{ij}}{(A_i + A_j) m_u},$$ we can write

$$\epsilon_{ij} = \frac{q_{ij}}{(1+\delta_{ij}) A m_u} \rho X_i X_j \langle \sigma v \rangle$$

if we take $\langle \sigma v \rangle \propto T^{\nu}$, we get

$$\epsilon_{ij} = \epsilon_{0,ij} X_i X_j \rho T^{\nu}$$ where

$$\epsilon_{nuc} = \sum_{i,j} \epsilon_{ij}$$

remember $\frac{\partial l}{\partial m} = \epsilon_{nuc} - \epsilon_{\nu} + \epsilon_{gr}$? this is that ϵ_{nuc}

The composition also changes:

$$\left(\frac{dn_i}{dt}\right)_j = -(1+\delta_{ij}) r_{ij} = -n_i n_j \langle \sigma v \rangle_{ij}$$

* if $i = j$ a reaction consumes 2 nuclei!

The nuclear lifetime is then

$$\tau_{i,j} = \frac{n_i}{|(dn_i/dt)_j|} = \frac{1}{n_j \langle \sigma v \rangle_{ij}}$$

We need to account for the fact that different reactions can both consume particles as well as produce particles. Eg. a reaction between particles k & l could produce particle i, so the total change is

$$\frac{dn_i}{dt} = -\sum_j (1+\delta_{ij})\, r_{ij} + \sum_{k,l} r_{kl,i}$$

If $n_i = X_i \rho / A_i m_H$, we can rewrite this as

$$\frac{dX_i}{dt} = \frac{A_i m_H}{\rho}\left(-\sum_j (1+\delta_{ij})\, r_{ij} + \sum_{kl} r_{kl,i}\right)$$

$$r_{ij} = \frac{\epsilon_{ij}}{q_{ij}(A_i+A_j)}\frac{\rho}{m_H}$$

The main nuclear burning cycles:

If you want to keep track of "everything" you need to calculate huge nuclear reaction tables. this is what is done for detailed stellar evolution codes!

If you want a general overview of structure and evolution, things can be simplified

* This is because

(1) nuclear fusion of different particles occurs at distinct temperatures because of the strong T dependence of the nuclear reaction rates and the importance of the Coulomb barrier

$\longrightarrow$ this leads to nuclear burning cycles

(2) in each burning cycle, only a few reactions dominate the energy production

(3) The rate of an entire reaction chain is limited by the slowest reaction. This is the one that dominates, so this is the one we pay attention to.

<u>Cycle 1: Hydrogen burning:</u>

$$4\,{}^1H \longrightarrow {}^4He + 2e^+ + 2\nu$$

In this reaction, two protons must be converted into neutrons which occurs from

$p \longrightarrow n + e^+ + \nu$, which is a weak interaction

* the 2ν's emitted are assumed to escape and only contribute to heating the gas of the star on the way out

A simultaneous reaction of 4 protons is unlikely. Instead, a chain of reactions leads to the production of ^4He.

P-P chains:

$$^1H + {}^1H \longrightarrow {}^2H + e^+ + \nu \quad \text{or} \quad p + p \rightarrow D + e^+ + \nu$$

* requires simultaneous β decay of one of the protons during the strong interaction
  $\Rightarrow$ very small cross section! $\sim 10^{-20}$ x's that of a typical strong interaction
  * never measured and only known theoretically because we see Deuterium

Once we have deuterium though, it will rapidly interact with another proton to produce ^3He

This can then lead to 3 possible branches:

$$^1H + {}^1H \rightarrow {}^2He + e^+ + \nu \; ; \; {}^2He + {}^1H \rightarrow {}^3He + \gamma$$

$$^3He + {}^3He \rightarrow {}^4He + 2\,{}^1H$$

$\boxed{\text{PP 1}}$

$$^3He + {}^4He \longrightarrow {}^7Be + \gamma$$

$$^7Be + e^- \rightarrow {}^7Li + \nu$$

$$^7Li + {}^1H \longrightarrow {}^4He + {}^4He$$

$\boxed{\text{PP2}}$

$$^7Be + {}^1H \rightarrow {}^8B + \gamma$$

$$^8B \rightarrow {}^8Be + e^+ + \nu$$

$$^8Be \rightarrow {}^4He + {}^4He$$

$\boxed{\text{PP3}}$

pp1: only the first two reactions are needed & have to occur twice.

pp2, pp3: only 1 occurence of first two reactions and a ^4He (primordial of previous burning).

* branching between 2 & 3 occurs when either an e^- is captured or a proton is fused.

$\langle E_\nu \rangle_{p-p} = 0.265\,MeV$; $\langle E_\nu \rangle_{^7Be+e^-} = 0.814\,MeV$

$\langle E_\nu \rangle_{^8B} = 6.71\,MeV$
Beta decay

→ $Q_{H,pp1} = 26.20\,MeV$
$Q_{H,pp2} = 25.66\,MeV$
$Q_{H,pp3} = 19.76\,MeV$

pp1 dominates over pp2, pp3 at $T \lesssim 1.5e7\,K$ because $^3He + ^3He$ is less sensitive to temperature than $^3He + ^4He$

→ pp1 dominates for sun-like stars.

as T increases, so do pp2, pp3.

- at low ($T < 8e6K$) temperatures, the rates of each reaction in the chain are low and important.

- at $T \gtrsim 8e6$ K, all reactions that happen after the slow production of D (or ^2H) are fast to produce ^4He

$\rightarrow$ D, ^3He, ^7Li, etc are short lived and thus their abundance is small.

* rate of reaction is due to small cross section, not # of interactions!

Because p+p reaction is slowest, the energy generation rate, $\epsilon_{nuc} = \sum_{i,j} \epsilon_{ij}$, is just

$$\epsilon_{nuc_{pp2,\,pp3}} = \frac{Q_H \, r_{pp}}{\rho}$$, where Q_H is the energy of the whole chain.

For pp1 : $\epsilon_{nuc_{pp1}} = \frac{1}{2} \epsilon_{nuc_{pp2}} = \frac{1}{2} \frac{Q_H \, r_{pp}}{\rho}$

We can sub in $q_H = \frac{Q_H}{4 m_u}$ and $r_{pp} = X^2 \rho^2 \langle \sigma v \rangle_{pp}$ to get

$$\epsilon_{nuc} = \psi \, q_H \, X^2 \frac{\rho}{m_u} \langle \sigma v \rangle_{pp}$$

ψ gives relative rate of pp1 & pp2, pp3 & is between 1 & 2

CNO cycle: if some C, N, O is present and T is high enough, you can also get H fusion through CNO

F: Flourine
Be: Berillium
B: Boron

C–N cycle

$$^{12}C + {}^{1}H \longrightarrow {}^{13}N + \gamma$$
$$^{13}N \longrightarrow {}^{13}C + e^{+} + \nu$$
$$^{13}C + {}^{1}H \longrightarrow {}^{14}N + \gamma$$
$$^{14}N + {}^{1}H \longrightarrow {}^{15}O + \gamma$$
$$^{15}O \longrightarrow {}^{15}N + e^{+} + \nu$$
$$^{15}N + {}^{1}H \longrightarrow {}^{12}C + {}^{4}He$$

or

$\longrightarrow {}^{16}O$ $\sim 10^{-3}$ x's less likely than $^{12}C + {}^{4}He$

$$^{16}O + {}^{1}H \longrightarrow {}^{17}F + \gamma$$
$$^{17}F \longrightarrow {}^{17}O + e^{+} + \nu$$
$$^{17}O + {}^{1}H \longrightarrow {}^{14}N + {}^{4}He$$

* rate depends on relative abundance of C, N, O
* total # of CNO nuclei is conserved

* β-decay happens on $\sim$ 100 – 1000 sec timescales
  $\longrightarrow$ ^{13}N, ^{15}O, ^{17}F have low abundance.

At high enough T, T > 1.5e7 K, the CNO cycle reaches a steady state & the rate is limited by the slowest interaction: $^{14}N(p,\gamma)^{15}O$ $\longrightarrow$ ^{14}N is most abundant

In a steady state, the $\frac{dn}{dt}$'s for each nucleon are equal, so we can write:

$$\left[\frac{n(^{12}C)}{n(^{13}C)}\right]_{steady} = \frac{\langle \sigma v \rangle_{13}}{\langle \sigma v \rangle_{12}} = \frac{\tau_p(^{12}C)}{\tau_p(^{13}C)}$$

where τ_p is the lifetime the nucleon exists without a proton capture.

$$\tau_p(^{15}N) \ll \tau_p(^{13}C) < \tau_p(^{12}C) \ll \tau_p(^{14}N) \ll \tau_{nuc}$$

$$\longrightarrow \epsilon_{CNO} = q_H X X_{14} \frac{\rho}{m_u} \langle \sigma v \rangle_{pN}$$

$\langle \sigma v \rangle_{pN}$ is cross section for $^{14}N(p,\gamma)^{15}O$

X: hydrogen abundance

X_{14}: ^{14}N abundance $\sim X_{CNO}$

q_H: $\frac{Q_H}{4m_u}$; $Q_H = 24.97$ MeV

CNO cycle is extremely temperature dependent:

$\epsilon_{pp} \propto X^2 \rho T^4$ $\quad$ $\epsilon_{CNO} \propto X X_{14} \rho T^{18}$ $\quad$ show fig 6.5

$\downarrow$ $\quad$ $\downarrow$

dominates @ lower T $\quad$ dominates @ higher T

Helium Burning:

occurs at higher temperatures because

(1) Coulomb barrier is higher

(2) Fusion of $^4He + ^4He$ requires two steps due to no stable A=8 nucleus

$$^4He + ^4He \longrightarrow ^8Be$$

$$^8Be + ^4He \longrightarrow ^{12}C^* \longrightarrow ^{12}C + \gamma$$

* ^8Be decays within 1e-16s

$\longrightarrow$ concentration increases at $T \approx 10^8 K$ and allows 2nd reaction to occur at appreciable rates due to Gamow peak.

This is called: triple-α reaction:

$$3\,^4He \longrightarrow ^{12}C + \gamma \quad : \quad Q = 7.25 \text{ MeV}$$

$\frac{1}{10}$th the amount released per unit mass compared to Hydrogen

Once you have enough ^{12}C, you can produce ^{16}O from

$$^{12}C + ^4He \longrightarrow ^{16}O + \gamma$$

Carbon burning and beyond:

Carbon burns at $T > 5\times10^8$ K

Neon burns at $T > 1.5\times10^9$ K

Oxygen burns at $T > 2e9$ K

Silicon burns at $T > 3e9$ K

@ $T > 4e9$ K: nuclear statistical equilibrium is reached where the most abundant nuclei are those with the lowest binding energy

$\longrightarrow {}^{56}Fe$

Skip Neutrino for now